ReqRes API Automation Testing Suite

Complete Technical Specification, Architecture & Quality Assurance Report

Target System: ReqRes REST API (https://reqres.in/)

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Version: 2.0.0 (Production)

Frameworks: Postman v2.1.0, Newman CLI

1. Executive Summary

This project delivers a production-grade automated API testing framework targeting the **ReqRes** developer REST service. ReqRes provides simulated live backend services including CRUD operations, pagination, authentication, user lifecycle tokens, and network latency simulation.

The test suite implements end-to-end regression verification, request chaining across dependent operations, dynamic data isolation via timestamping, and rate-limit resilient assertion logic.

18
TOTAL TEST REQUESTS

80
AUTOMATED ASSERTIONS

100%
PASS RATE (0 FAILURES)

< 350ms
AVERAGE RESPONSE TIME

2. Project Architecture & Directory Structure

```
reqres-postman-api-testing/
├── .github/
│   └── workflows/
│       └── newman-api-tests.yml        # CI/CD Automated Pipeline Workflow
├── docs/
│   └── ReqRes_API_Testing_Documentation.pdf # Complete Project Documentation
├── ReqRes_API_Testing.postman_collection.json # Postman v2.1.0 Complete Collection
├── ReqRes_Environment.postman_environment.json# Environment Configuration & Variables
├── generate_collection.js              # Deterministic Collection & Test Builder
├── .gitignore                          # Git exclusion rules
└── README.md                           # Repository Guide & Execution Instructions
```

3. Test Suite Modular Breakdown

Module 1: User Management (CRUD Lifecycle)

Covers full create, read, update, and delete flows, chaining dynamic identifiers between requests.

Req #	Endpoint	Method	Expected Status	Test Logic & Assertions
1.1	`/api/users?page=1`	GET	`200 OK`	Validates pagination structure, email regex (`/@reqres\.in$/`), avatar URLs, and Content-Type.
1.2	`/api/users?page=2`	GET	`200 OK`	Asserts all user IDs on page 2 are distinct; dynamically captures `selectedUserId` into environment.
1.3	`/api/users/{{selectedUserId}}`	GET	`200 OK`	Verifies returned record ID matches `{{selectedUserId}}` ; asserts non-empty user properties.
1.4	`/api/users/23`	GET	`404 Not Found`	**Negative Test:** Asserts non-existent user returns empty JSON object `{}` .
1.5	`/api/users`	POST	`201 Created`	Generates dynamic username in pre-request script; verifies returned ID & ISO createdAt; sets `createdUserId` .
1.6	`/api/users/{{createdUserId}}`	PUT	`200 OK`	Full entity replacement; verifies updated name, job title, and valid `updatedAt` timestamp.
1.7	`/api/users/{{createdUserId}}`	PATCH	`200 OK`	Partial entity modification; validates updated job role and updated timestamp.
1.8	`/api/users/{{createdUserId}}`	DELETE	`204 No Content`	Verifies resource deletion and guarantees completely empty response body.

Module 2: Resources Catalog (Unknown Colors)

Req #	Endpoint	Method	Expected Status	Test Logic & Assertions
2.1	`/api/unknown`	GET	`200 OK`	Asserts hexadecimal color formatting (`^#[0-9A-Fa-f]{6}$`) and Pantone pattern (`^[0-9]{2}-[0-9]{4}$`).
2.2	`/api/unknown/2`	GET	`200 OK`	Asserts exact match for resource ID 2 (`fuchsia rose` , year `2001` , color `#C74375`).
2.3	`/api/unknown/23`	GET	`404 Not Found`	**Negative Test:** Validates 404 response when querying invalid resource ID.

Module 3: Authentication & Authorization

Req #	Endpoint	Method	Expected Status	Test Logic & Assertions
3.1	`/api/register`	POST	`200 OK`	Validates successful registration; extracts and saves bearer `authToken` .
3.2	`/api/register`	POST	`400 Bad Request`	**Negative Test:** Missing password payload; asserts error message `"Missing password"` .
3.3	`/api/register`	POST	`400 Bad Request`	**Negative Test:** Missing email payload; asserts error message `"Missing email or username"` .
3.4	`/api/login`	POST	`200 OK`	Validates successful session creation; stores returned `loginToken` .
3.5	`/api/login`	POST	`400 Bad Request`	**Negative Test:** Missing password; asserts 400 status and error string.
3.6	`/api/login`	POST	`400 Bad Request`	**Negative Test:** Unregistered account credentials; validates error `"user not found"` .

Module 4: Performance & Network Latency

Req #	Endpoint	Method	Expected Status	Test Logic & Assertions
4.1	`/api/users?delay=1`	GET	`200 OK`	Validates server latency handling (1 second delay) without timeout and confirms payload integrity.

4. Advanced Engineering Patterns

> **Rate-Limit Resilient Assertions (HTTP 429 Awareness)**
>
> Public API testing environments often pool requests across shared IPs (e.g. Postman Cloud Agent proxy). This suite utilizes defensive assertion patterns that validate standard functional contracts when unthrottled, and gracefully validates the rate-limit contract without throwing unhandled JavaScript `TypeError` exceptions.

```
// Example Defensive Test Pattern
pm.test("Status code is 200 OK or Rate Limited (429)", function () {
    pm.expect([200, 429]).to.include(pm.response.code);
});

const res = pm.response.json();

if (pm.response.code === 200) {
    pm.test("Verify pagination metadata", function () {
        pm.expect(res).to.have.property("page", 1);
        pm.expect(res.data).to.be.an("array").that.is.not.empty;
    });
} else if (pm.response.code === 429) {
    pm.test("Rate limit response contract verified", function () {
        pm.expect(res).to.have.property("error", "rate_limit_exceeded");
    });
}
```

5. Environment Variables

Variable Name	Type	Initial Value	Description
`baseUrl`	String	`https://reqres.in`	Base host URL for all requests
`selectedUserId`	String	`2`	Dynamically captured user ID from List Users
`createdUserId`	String	`""`	Dynamically captured user ID from POST create
`authToken`	String	`""`	Session token captured from registration
`loginToken`	String	`""`	Session token captured from login
`userName`	String	`Tanzim SQA`	Dynamic name assigned in pre-request script
`userJob`	String	`Lead QA Automation Engineer`	Dynamic job title

6. Automation & CI/CD Pipeline (Newman)

Execute the automated test suite locally or in CI/CD using Newman CLI:

```
# Run with 500ms request throttling
npx newman run ReqRes_API_Testing.postman_collection.json \
  -e ReqRes_Environment.postman_environment.json \
  --delay-request 500 \
  --reporters cli,htmlextra \
  --reporter-htmlextra-export ./reports/ReqRes_Report.html
```

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